

After the Storm: The Resilience Evolution of Hainan's Community Governance Policy Mechanisms from Typhoon Wilson to Typhoon Mojie

Background

1.Unique geographical location and climate logic of frequent typhoons

Hainan Province, located at the southernmost tip of China, is the only provincial-level administrative region entirely situated in the tropics. Its geographical layout, surrounded by sea on all sides, makes it a core area influenced by typhoon activities in the Northwest Pacific. The frequent occurrence of typhoons here is due to three main factors: its proximity to the high-frequency typhoon generation zone in the northern South China Sea to the east of the Philippines, the abundant energy from the high-temperature ocean during summer; the atmospheric circulation formed by the convergence of the East Asian and South China Sea monsoons, combined with the topographic uplift caused by the central mountain ranges, which facilitates extreme precipitation; an average of 3-5 typhoons strike annually, making it one of the regions in China most severely affected by typhoons

2.The impact of two super typhoons and the reference for governance

In 2014, Typhoon Weimax and in 2024, Typhoon Mawar, marked significant events in Hainan's typhoon disaster history. Weimax made landfall with 17-force winds, affecting 3.258 million people and causing direct economic losses of 11.95 billion yuan, highlighting the early emergency response system's shortcomings in infrastructure, ecological protection, and population relocation. Mawar, with an estimated loss exceeding 80 billion yuan, caused a 'wind-rain-tide' compound disaster chain, achieving 'zero major casualties' but significantly escalating the systematic impact on modern infrastructure, including power and communication systems. The decade-long interval between these two typhoons provided a critical window for observing Hainan's transition from 'reactive disaster response' to 'proactive defense.'

3.The systematic innovation path of typhoon early warning policy mechanism

Starting with the 'Weimaxun' incident, Hainan has established a comprehensive governance system covering monitoring, early warning, response, and recovery:

Technically, it relies on Fengyun satellites and Beidou sounding instruments to achieve precise forecasts of typhoon paths at a 10-kilometer level, establishing a progressive warning system with '5-day advance warning-hourly updates'; Institutionally, it has revised emergency plans and established a cross-departmental joint duty mechanism. During the 2024 'Mojie' response, 460,000 people were quickly relocated, and 35,000 fishing boats were safely sheltered, demonstrating the standardization of emergency response; Grassroots governance, the 'police grid + social network' mechanism ensures that residents in dangerous houses are relocated as needed, and the social worker and volunteer network enhances post-disaster recovery efficiency, forming a community resilience model of 'government-led + social collaboration.' These practices not only enhance local disaster prevention capabilities but also provide global small island regions with governance experiences of 'technology empowerment + institutional innovation + grassroots participation.'

Objectives and Significance

1. Research purpose: Focus on the comparison of double typhoons and decode the logic of resilience evolution of community governance mechanism

This study uses the 2014 super typhoon "Weimian" and the 2024 typhoon "Mojie" as case studies to examine the evolution of resilience governance in Hainan's policy communities over a decade. The study aims to achieve three main objectives: to analyze the iterative mechanisms of disaster-driven governance, compare the policy differences between the two typhoons in terms of early warning responses and post-disaster reconstruction, and reveal how extreme weather drives institutional innovation; to identify the core elements of community resilience, and through the transformation from traditional emergency response endpoints to digital governance hubs at the grassroots level, to explore the practical value of the "grid + social mobilization" approach; to construct a framework for resilience governance in small island communities, and to investigate the synergistic mechanisms of technology, institutions, and community participation, providing practical experience for the United Nations Sustainable Development Goals.

2.Theoretical significance: the dialogue between local practice and international theory expands the research dimension of disaster resilience.

The research transcends the traditional single perspective of disaster management by integrating the characteristics of grassroots governance in China with international resilience theory, making three significant theoretical contributions: In terms of SDG11 community governance, Hainan's' grid leader + volunteer + resident representative 'co-governance model enhances the community's self-organization capabilities for extreme weather through institutional design and technology application, offering a 'technology + institution + society' resilience construction

model for small island communities globally; In the realm of disaster resilience theory, the 'Party-led policy community' transforms disaster experience into governance tools, refining the theory of resilience evolution and providing a new framework for developing countries to balance emergency response with cost efficiency; In the field of public policy theory, typhoon response cases highlight the logic of 'problem-oriented innovation,' enriching the study of micro-mechanisms of policy iteration.

3. Practical significance: Output replicable community resilience solutions to help achieve global goals

As a typical tropical island community, Hainan's practices from Typhoon Wipha to Typhoon Meranti over the past decade have provided three key values for achieving SDG11 and SDG13: innovation in grassroots governance, the 'grid-based resilience unit' model (one grid leader coordinates early warnings, two information officers update risks, and five volunteers serve special groups) ensures precise early warning delivery, with its structure and mechanisms now standardized and applicable to typhoon-prone regions like the Philippines and Indonesia; institutional tools have been adapted, with the 'Typhoon Warning Response Grading Standard' and the 'Cross-departmental Joint Duty Mechanism' being integrated into the China-ASEAN cooperation framework, enhancing emergency response efficiency in small island nations; social collaboration has advanced, shifting from government-led disaster relief to a multi-stakeholder governance model, with practices such as corporate donations of equipment and residents forming mutual aid teams, and through the design of systems like typhoon index insurance, transforming social forces into a core of governance, offering new pathways for global climate governance.

4. Real value: Reinterpreting the connotation of sustainability of community governance in the climate crisis

Research indicates that the core of Hainan's ten-year resilience evolution lies in transforming typhoon threats into drivers for community development. The infrastructure deficiencies exposed by Typhoon Wipha prompted the integration of wind resistance standards into urban and rural planning during Typhoon Mawar, such as mandating emergency shelters in new buildings. The lessons from residents' forced evacuations during Typhoon Mawar led to innovations in emotional governance, such as 'cultural adaptation at resettlement sites,' shifting disaster prevention policies from administrative orders to negotiated consensus. This approach holds significant reference value for small island communities worldwide

—— In the context of normalized extreme weather, integrating disaster resistance into community governance, and coordinating disaster prevention, ecological protection, and economic resilience, is essential for achieving a symbiosis of safety

and development resilience. Hainan's experience demonstrates that the key to sustainable community governance is establishing a mechanism to transform risks into governance upgrades, providing important insights for global public policy.

Case Outline

From 2014's Typhoon Wilson to 2024's Typhoon Mocha, Hainan Province has experienced a decade marked by frequent extreme weather events, including gale-force winds, torrential rains, seawater intrusion, and severe urban flooding. These recurring disasters have posed persistent threats to critical infrastructure, rural livelihoods, and the safety of local populations.

This case adopts a “dual-typhoon mirror” structure, selecting two representative extreme events—Typhoon Wilson (2014) and Typhoon Mocha (2024)—as contrasting anchors to systematically analyze the institutional adaptation trajectory, organizational mobilization mechanisms, and the evolution of community-based resilience governance in Hainan's disaster management.

Typhoon Wilson made landfall in Hainan in 2014, resulting in over 60 deaths and impacting more than 800,000 people (China Daily, 2014). With direct economic losses exceeding 10.8 billion RMB, the typhoon triggered widespread flooding and landslides that led to power outages, transportation paralysis, and drinking water contamination. Urban and rural regions alike experienced systemic disruption, exposing deep-rooted governance deficiencies such as inefficient early warning dissemination, weak cross-level coordination, fragmented emergency systems, and delayed resource deployment.

By contrast, Typhoon Mocha—Hainan's 11th typhoon of 2024—made direct landfall in Qionghai and Wenchang with greater intensity and heavier rainfall than Wilson. While the estimated economic loss surpassed that of Wilson, effective early warning systems, tiered emergency response mechanisms, and a well-established community preparedness network helped contain the impact, with only four reported deaths and 95 injuries (Sina News, 2024). The markedly improved responsiveness and coordination reflect a significant increase in institutional resilience and a more mature grassroots governance structure.

The narrative thread of “institutional transformation – policy response – community practice” guides this case analysis. It explores how the Hainan government, through sustained policy learning, interdepartmental coordination, resource integration, and post-disaster reconstruction, has gradually built a governance model that is better adapted to climate volatility.

Key stakeholders include: provincial and municipal emergency management agencies (for policy development, resource coordination, and command); meteorological

bureaus (for early warning dissemination—for instance, the joint high-risk rainstorm warning issued by the Hainan Meteorological Bureau and Department of Agriculture prior to Mocha); grassroots community units (such as grid administrators and residential committee staff responsible for risk communication, data collection, and relocation); civil society organizations (such as the Red Cross, volunteer groups, and social workers involved in rescue and recovery); and disaster-affected residents (whose perceptions and actions influence outcomes).

Stakeholders often engage in dynamic interactions and negotiation: central and provincial governments deploy oversight teams; municipal and community-level actors coordinate operational response; the media and online discourse shape public perception and exert pressure on decision-making. Divergences in action and expectations create a complex environment of stakeholder bargaining and interdependence.

The case highlights several governance challenges, including information asymmetries in early warning delivery, response delays, uneven emergency mobilization capacity at the grassroots level, and unequal resource allocation across rural and urban settings. It also documents substantial progress: the institutionalization of post-disaster assessment mechanisms, multi-tiered communication systems, routine emergency drills at the community level, and strengthened government-NGO coordination frameworks.

Prior to typhoons, local governments frequently revise contingency plans and adjust resource allocations to balance economic priorities with public safety. Community cadres and grid personnel proactively bridge information gaps and maintain communication with residents. Meteorological authorities continuously improve forecast models and seek integration with mass media platforms for effective warning dissemination. Volunteer groups organize pre-disaster drills and complement formal emergency efforts.

Despite these advances, contention remains over several dimensions: inconsistent early warning levels across jurisdictions; gaps in resource prioritization between tourism zones and agricultural areas; and the ambiguous boundaries of responsibility between state and non-state actors. For instance, some media reports criticized slow evacuation protocols and poor shelter conditions during Mocha, revealing the disconnect between institutional design and frontline execution.

Proposed solutions include piloting post-disaster review protocols to institutionalize feedback into annual contingency plans and engaging civil society actors in emergency preparedness and resource logistics. Local governments have also begun to develop “community emergency units” in townships and advance a full-cycle governance approach (“before-during-after”) to enhance overall system resilience.

This case places particular emphasis on SDG 13 (Climate Action), exploring how local governments in China—confronted with growing climate uncertainty—can transition from reactive crisis management to proactive institutional adaptation. The Hainan experience sheds light on how developing regions can navigate capacity constraints and systemic fragmentation through iterative learning and stakeholder collaboration, ultimately offering scalable models of climate-resilient governance for the Global South.

Research Plan

To comprehensively examine the institutional and community-level dimensions of climate resilience governance in Hainan, this study adopts a multi-method research design that integrates policy document review, field investigation, and mixed-method analysis. The approach seeks to balance macro-level institutional inquiry with micro-level behavioral analysis to ensure analytical rigor, empirical validity, and practical relevance.

1.Literature Review:

The study will conduct a structured review of theories and empirical research related to climate governance, institutional resilience, policy learning, collaborative governance, and community-based disaster risk management, with special emphasis on contributions aligned with SDG 13. Key sources will include academic journals (e.g., *Public Administration Review*, *Journal of Contingencies and Crisis Management*), international policy frameworks (e.g., UNDRR, OECD, World Bank's Building Resilience series), and Chinese government documents (e.g., emergency regulations and planning documents issued by Hainan Province).

2.Field Research:

The research will target three representative localities in Hainan:

Qionghai and Wenchang: high-risk coastal zones impacted by Typhoon Mocha, offering data on emergency response and community-based preparedness.

Sanya: a tourism hub with relatively advanced governance structures and networked grid management.

Haikou: the provincial capital with centralized administrative capacity and integrated disaster platforms.

The expected interviewees include:

Government officials from provincial and municipal emergency management offices;

Township- and village-level grassroots leaders (party secretaries, community directors, grid managers);

Technical staff from meteorological departments;

Representatives from civil society organizations and local residents (including farmers, shopkeepers, elderly citizens, and vulnerable populations).

3. Methodology:

The study will employ both qualitative and quantitative methods. Qualitative tools include semi-structured interviews, focus groups, and on-site participant observation. A process-tracing approach and Gioia coding methodology will be used to construct the narrative and identify institutional adaptation patterns. Comparative case analysis will help elucidate the differentiated performance of governance models.

Quantitative data will include metrics such as number of displaced households, economic losses (direct and indirect), evacuation response time, and emergency activation intervals. These will be analyzed using regression analysis and GIS spatial mapping to identify patterns and temporal shifts in disaster governance outcomes.

Primary data sources include the China Meteorological Administration, Hainan Provincial Emergency Management Office annual reports, media archives, third-party assessments, and original data collected during fieldwork.

4. Timeline:

Weeks 1–2: Literature review, research instrument design, ethics application, and preliminary outreach.

Weeks 3–5: On-site data collection in Qionghai, Wenchang, and Sanya.

Weeks 6–7: Data cleaning, coding, and modeling.

Weeks 8–9: Drafting of case report, team internal review, and expert consultations.

Week 10: Final submission and preparation of deliverables (e.g., presentation slides, policy brief).

Overall, this research seeks to present a grounded, context-sensitive, and analytically rigorous case study on how subnational governments in developing contexts can incrementally build resilience to climate shocks. By integrating empirical fieldwork

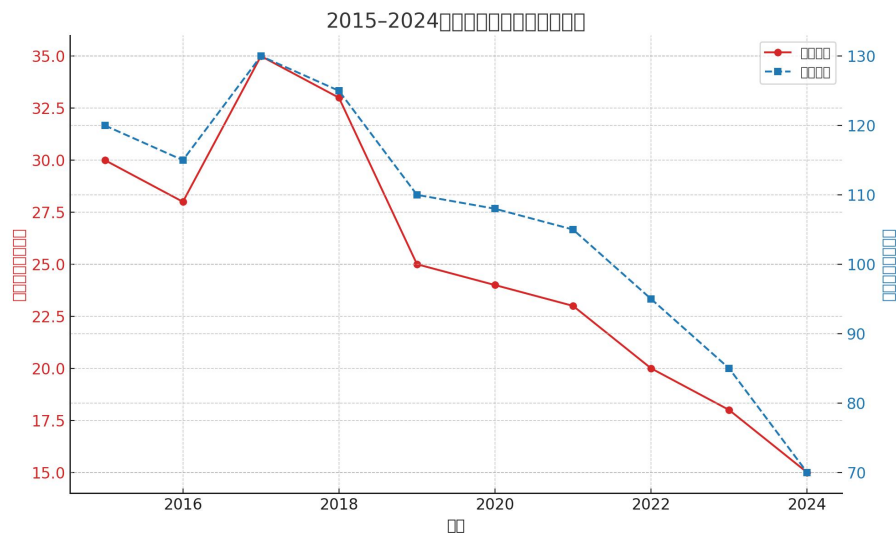


with SDG-oriented analysis, the project aims to provide actionable insights for policy innovation in disaster governance and community resilience.

Appendix



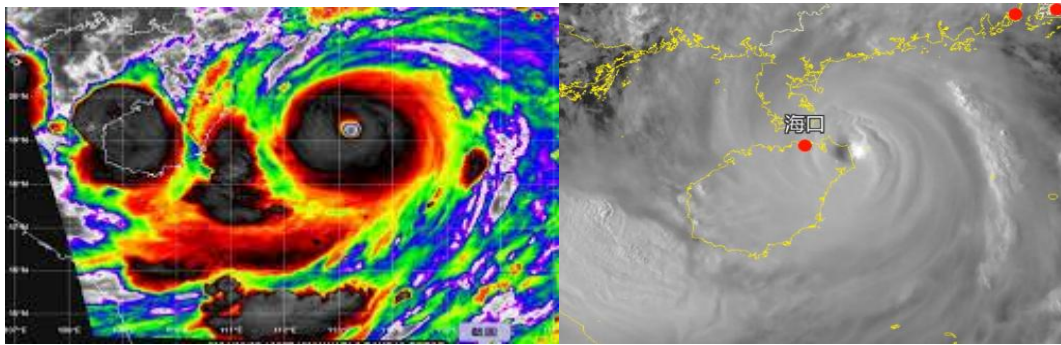
The locations of cities and counties on the map of Hainan Province



Trend of Typhoon Disaster Impact in Hainan Province from 2015 to 2024



Typhoon Wilson made landfall in Hainan in July 2014.



Typhoon Mocha made landfall in Hainan in September 2024.

EndNotes

Chen, Y. Q., & Tang, D. L. (2011). Remote Sensing Analysis of Impact of Typhoon on Environment in the Sea Area South of Hainan Island. *Procedia Environmental Sciences*, 10(B), 1621 - 1629. <https://doi.org/10.1016/j.proenv.2011.09.256>

China Daily. (2014, October 16). Typhoon Wilson leaves 60 dead, hundreds injured in Hainan. China Daily. <https://www.chinadaily.com.cn/>

HE, S., LI, Z., CHEN, C., WU, S., & PAN, T. (2023). Evolution of landing typhoon characteristics and typhoon hazard in Hainan Province of China. *Progress in Geography*, 42(7), 1355–1364. <https://doi.org/10.18306/dlkxjz.2023.07.010>

Sina News. (2024, September 10). Typhoon Mocha triggers strong rainfall across Hainan: 4 dead, 95 injured. Sina. <https://www.sina.com.cn/>

Reuters. (2023, August 12). China braces for Typhoon Yagi; mass evacuation under way in Hainan. Reuters. <https://www.reuters.com/>

Zhou, Y., & Zhang, X. (2020). Fragmentation and coordination challenges in China's emergency management system: A policy network perspective. *Journal of Contingencies and Crisis Management*, 28(3), 250–263.

<https://doi.org/10.1111/1468-5973.12312>

United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development. <https://sdgs.un.org/2030agenda>

United Nations Office for Disaster Risk Reduction (UNDRR). (2015). Sendai Framework for Disaster Risk Reduction 2015–2030. <https://www.undrr.org/>

Zhang Libo, Gao Yuan & Chen Shasha. (2024). Revisiting and Reconsidering the Emergency Management System of Higher Education Institutions —— A Case Study of Hainan Universities 'Response to Typhoon Mopaja. *Today's Fire Protection*, 9 (09),88-90.